

David Gómez Agudelo

Medellín, Antioquia, Colombia • David.gomez.agudelo@gmail.com • (+57) 3216455813 • LinkedIn • GitHub

Professional Profile

Ninth-semester Systems Engineering student and Software Development Technician, passionate about software development, quality assurance, and data analysis, with a strong interest in creating technology solutions that add value to business processes. I have experience developing web applications, software testing, and data management using Java, Spring Boot, PHP, Laravel, React, Next.js, SQL, REST APIs, and Data Science tools. I have participated in healthcare electronic invoicing, document management, and process automation projects, with a focus on best practices, security, quality, performance, and software design.

Experience

Universidad de Antioquia

Medellín, Colombia

Programming Assistant

March 2026 – Present

- Developed and implemented ARCA, a Laravel-based web application for document management for teaching applicants, integrating authentication through an institutional API. The system eliminated approximately 50 daily emails that were previously sent directly to the assigned assistant, centralizing the process into a single workflow with full traceability: applicants upload their documents, the Academic Affairs department reviews them and records observations, and Human Resources performs the final validation and archiving.
- Develop and maintain internal tools and academic applications using PHP and MySQL, ensuring the stability and proper operation of existing systems.
- Optimize complex SQL queries and write detailed technical documentation, improving system performance and facilitating knowledge transfer within the team.
- Collaborate directly with the IT team on the integration of external APIs, promoting service scalability across university systems.

Selected Projects

Healthcare Electronic Invoicing System

Full-Stack / RPA Development

Next.js, Spring Boot, PostgreSQL, Python, Selenium, Supabase

Sep. 2025 – Feb. 2026

- Designed and implemented the Invoicing, Admissions, and Administration modules of a multi-team institutional platform, integrating business processes with DIAN and Ministry of Health requirements for electronic invoicing and RIPS reporting.
- Developed the electronic invoicing workflow, enabling invoice creation, generation of required documents, and submission for validation to DIAN, as well as the generation and reporting of RIPS information to the Ministry of Health.
- Implemented an RPA component using Selenium to automate the submission of supporting documents to healthcare payer entities, incorporating a hybrid workflow for portals with CAPTCHA and automatically generating organized and renamed document packages.
- Designed the Admissions module as a cross-functional component to centralize patients, CUPS codes, authorizations, and roles, reducing information fragmentation between Medical Records, Medical Appointments, and Billing teams.

School Management System

Full-Stack Development

React, Spring Boot, Spring Data JPA, MySQL

- Designed and developed a comprehensive application for school administration, covering complete modules for users, groups, subjects, and grades.

- Implemented CRUD operations and complex relationship mapping using JPA and Hibernate, with centralized exception handling and strict endpoint validation.
- Integrated secure authentication using Google OAuth2 and role-based access control for Administrators, Teachers, and Students.

Popular News Prediction

Machine Learning / Data Science

Python, scikit-learn, Pandas, Matplotlib

- Trained a classification model to predict news popularity by analyzing titles, categories, publication days, and keywords, building a reproducible pipeline in Jupyter Notebook with complete exploratory data analysis.
- Evaluated model performance using key statistical metrics, including Accuracy, Precision, Recall, and confusion matrices.

Web Novels Platform

Full-Stack Development

TypeScript, Next.js, PostgreSQL, Tailwind CSS

- Developed a full-stack platform for exploring and importing books, featuring secure JWT authentication with automatic token rotation, a scraping engine for automatically importing stories from different platforms, and an administration panel with statistics and user, tag, and fandom management.
- Designed a modern and fully responsive interface using Next.js and Tailwind CSS.

Technical Skills

Programming Languages: Java, PHP, Python, JavaScript, SQL.

Frameworks and Libraries: Laravel, Spring Boot, Next.js, React, Hibernate, Pandas, NumPy, scikit-learn, Tailwind CSS.

Tools and Databases: Git, Docker, MySQL, PostgreSQL, Supabase, REST/SOAP APIs, Selenium/RPA.

Methodologies and Concepts: Hexagonal Architecture, Unit and Integration Testing, MVC Patterns, SOLID Principles, Agile Development (Scrum), Data Science.

Education

Universidad de Antioquia

Medellín, Colombia

Systems Engineering (Ninth-Semester Student)

2022 – Present

Relevant Coursework: Data Structures, Databases, Requirements Engineering, Software Quality.

CESDE

Medellín, Colombia

Software Development Technician

2020 – 2021

Academic Focus: Web and desktop application development, Object-Oriented Programming fundamentals, and Relational Databases.